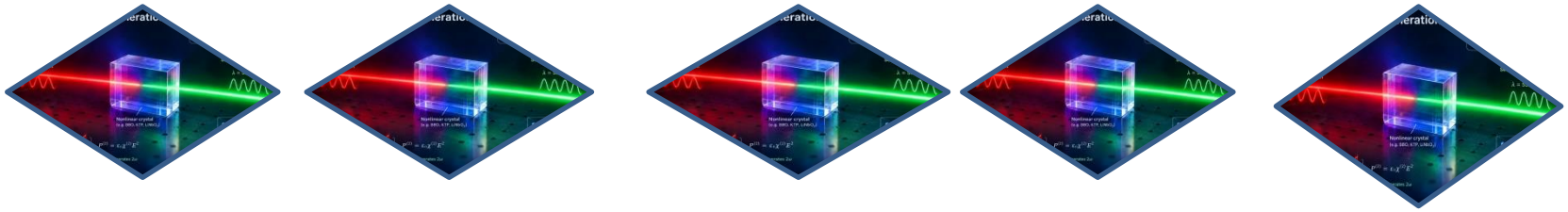
Nonlinear Optics



Lecture 3:Second Harmonic Generation (SHG)

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Second Harmonic Generation (SHG)

Introduction :

Second Harmonic Generation (SHG) is a nonlinear optical process in which two photons of the same frequency interact with a nonlinear optical medium to produce a photon with twice the original frequency.

If the incident light has angular frequency ω, the Generated light has angular frequency 2ω.

$$\omega + \omega = 2\omega$$

This process is also known as frequency doubling.

Physical Principle of SHG

In a linear optical medium, the induced polarization is proportional to the applied electric field:

$$P = \varepsilon_0 X^{(1)} E$$

In a nonlinear optical medium, the polarization can be expressed as:

$$P = \varepsilon_0 [\chi(1)E + \chi(2)E2 + \chi(3)E3 + \cdots]$$

The second-order nonlinear polarization is:

$$P^{(2)} = \varepsilon_0 \mathbf{X^{(2)}} E2$$

Where:

$P^{(2)}$: Second-order nonlinear polarization.

ε_0 : Permittivity of free space.

$\mathbf{X^{(2}}$: Second-order nonlinear susceptibility.

E : Electric field of the incident light.

For an incident electric field:

$$\mathbf{E(t) = E_0 \cos(\omega t)}$$

Substituting into the second-order polarization equation gives:

$$\mathbf{P^{(2)} = \varepsilon_0 \chi^{(2)} E_0^2 \cos^2(\omega t)}$$

Using the trigonometric identity:

$$\mathbf{\cos^2(\omega t) = \tfrac{1}{2}[1 + \cos(2\omega t)]}$$

Therefore:

$$\mathbf{P^{(2)} = (\varepsilon_0 \chi^{(2)} E_0^2 / 2)[1 + \cos(2\omega t)]}$$

This equation shows that the nonlinear polarization contains a component oscillating at 2ω, which can generate the second harmonic.

Frequency and Wavelength Relationships

The frequency of the second harmonic is twice the fundamental frequency:

$$\boxed{f_{2\omega} = 2f_{\omega}}$$

In vacuum, the wavelength of the second harmonic is half the fundamental wavelength:

$$\boxed{\lambda_{2\omega} = \frac{\lambda_{\omega}}{2}}$$

Where:

f_{ω} :Fundamental frequency.

$\mathbf{f_{2\omega}}$:Second-harmonic frequency.

λ_{ω}: Fundamental wavelength.

$\lambda_{\mathbf{2\omega}}$:Second-harmonic wavelength.

Phase Matching Condition

- Efficient SHG requires the fundamental wave and the generated second-harmonic wave to maintain a suitable phase relationship as they propagate through the nonlinear crystal.

$$\mathbf{k_{2\omega} = 2k_{\omega}}$$

- Since $k = n\omega/c$, the phase-matching condition can be written, for the selected propagation and polarization modes, as:

$$\mathbf{n_{2\omega} = n_{\omega}}$$

- Here, n is the refractive index and k is the wave number. In practice, birefringent crystals can be used to achieve phase matching. Phase matching improves the efficiency of energy transfer from the fundamental wave to the second harmonic.

Solved Examples

Example 1: A laser has a fundamental wavelength of 1064 nm. **Calculate the wavelength of the second harmonic.**

Solution:

$\lambda_{2\omega} = \lambda_{\omega} / 2$

$\lambda_{2\omega} = 1064 / 2 = 532$ nm **(visible green light).**

Example 2: The fundamental frequency of a laser is 4.5×10^{14} Hz. **Calculate the second harmonic frequency.**

Solution:

Given: $f\omega = 4.5 \times 10^{14}$ Hz

$$f_2\omega = 2f\omega$$

$f_{2\omega} = 2 \times (4.5 \times 10^{14}) =$ **9.0×10^{14} Hz**

Example 3:A laser has a fundamental frequency of 5×10^{14} Hz. **Calculate the wavelength of its second harmonic in vacuum.**

1. $f_2\omega = 2f\omega$

   $= 2 \times (5 \times 10^{14}) = 1 \times 10^{15}$ Hz

2. $c = 3 \times 10^8$ m/s

   $c = f\lambda$

   $\lambda_2\omega = c / f_2\omega$

$\lambda_2\omega = (3 \times 10^8) / (1 \times 10^{15})$

=300 nm

Factors Affecting SHG Efficiency

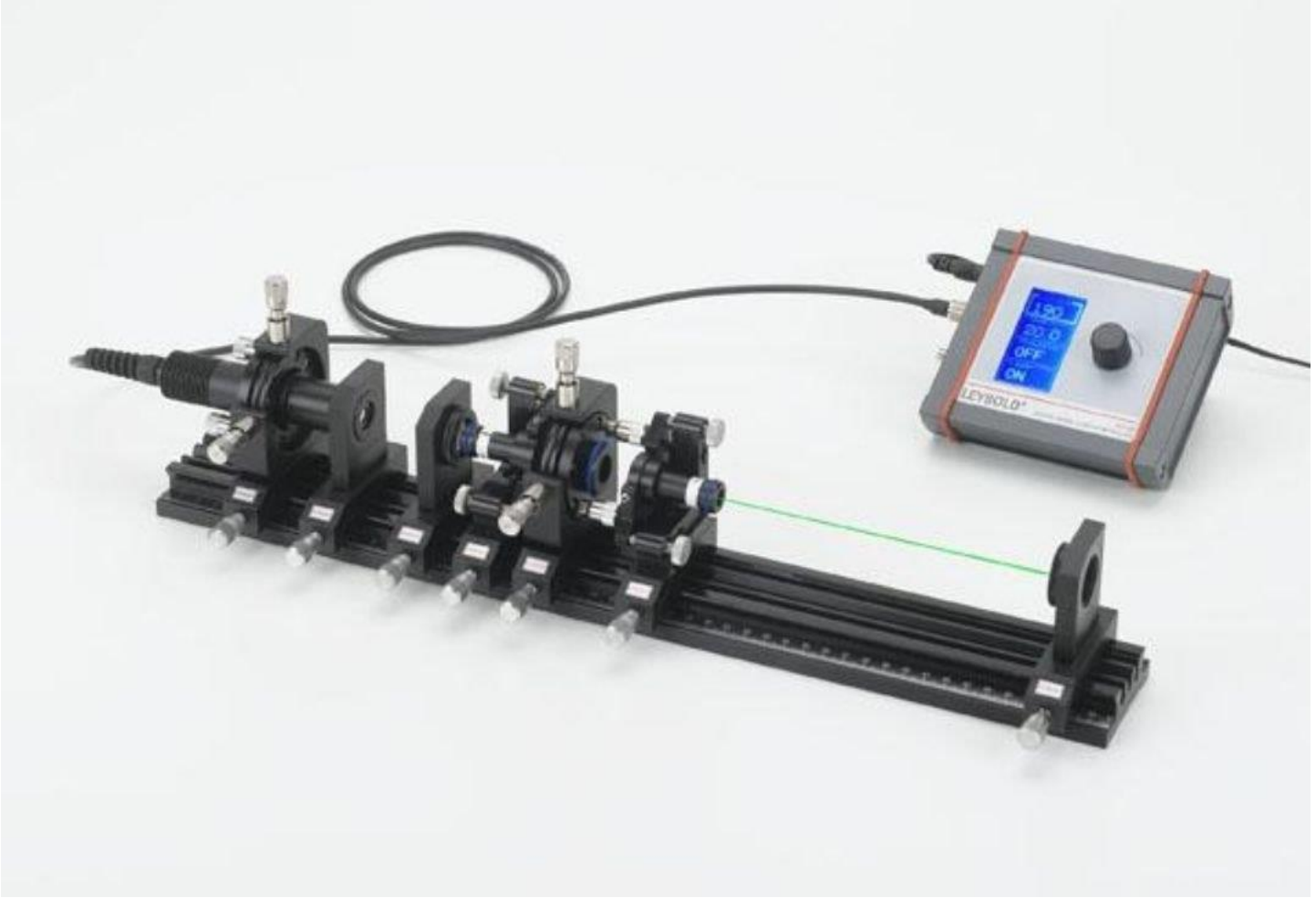
- Incident light intensity: higher intensity generally increases SHG output.
- Nonlinear susceptibility $\chi^{(2)}$: a larger effective nonlinear susceptibility can increase conversion efficiency.
- Crystal length: a longer crystal can improve conversion when phase matching is maintained.
- Phase matching: proper phase matching allows the generated waves to add constructively.
- Polarization and propagation direction: these determine the effective nonlinear response of the crystal.
- Optical absorption: absorption reduces the power available for conversion and transmission.

Applications of SHG

- **Laser frequency doubling:** converting infrared light at 1064 nm into visible green light at 532 nm.

- **Biomedical imaging:** SHG microscopy can image certain ordered biological structures without conventional fluorescent labels.

- **Optical frequency conversion:** producing new optical frequencies for laser systems, spectroscopy, and research.



Exercises for Students

Q1. A laser has a fundamental wavelength of 800 nm. Calculate the second harmonic wavelength.

Q2. The fundamental frequency of a laser is 3.5×10^{14} Hz. Calculate the second harmonic frequency.

Q3. Explain why phase matching is important in second harmonic generation.

Q4. Why are non-centrosymmetric crystals commonly used for second-order nonlinear optical processes?

